

THE ABUNDANCE LEDGER · PART 1 OF 2 · AI & ROBOTICS

Where Are the Robots?

When does robot labor reach mass usage, what does it do to productivity, why do two numbers nobody publishes decide both, and what would make the forecast trustworthy?

TON618 CAPITAL RESEARCH · SEC FILINGS THROUGH 26 AUGUST 2026 · BLS
PRODUCTIVITY REVISED 3 SEPTEMBER 2026 · COMPANY DISCLOSURES, CHINESE
EXCHANGE-FILING COVERAGE AND VENDOR PRICES AS OF 4 SEPTEMBER 2026

ONE · WHEN	
Robot labor at 1% of US hours worked, industry case / house base case	2028 / 2032
TWO · PRODUCTIVITY	
Added to annual productivity growth 2028–35, house base case	+0.25 pt
THREE · THE TWO NUMBERS	
Every delivered humanoid labor-hour any company has disclosed	66,000 h
FOUR · CONFIDENCE	
Disclosures the forecast needs, from any company in any country	0 of 5
BALANCE CARRIED FORWARD	price is not the gate; task success and supervision are

THE FOUR ANSWERS

When, how much, why two numbers decide it, and what we would need to know

This note set out to answer four questions that the argument about robots and abundance keeps circling without settling. When, realistically, do cost curves and production reach mass usage? What is the best guess for what that does to productivity? Why do two numbers that no robot company publishes, the share of tasks a robot completes and the number of humans it takes to watch it, decide both? And what would we need to know to trust the guess? Everything in the note is evidence for one of those four, and each section says which.

1. When: the price is already there, the volume is in China, and the hours are nowhere. Under the only filed cost structure in the industry, every published humanoid price from \$13,500 to \$200,000 produces a labor-hour cheaper than the \$24.51 US median wage. Price stopped being the constraint before the first price list. Volume exists: China built about 12,800 humanoids in 2025, about 90% of the world's, and Unitree's average price fell from ¥593,000 to ¥168,000 in two years. What does not exist, in China or the United States, is a single deployment that discloses how many hours a robot delivers. Three-quarters of Unitree's buyers are research and education customers and 3% to 4% of its units do operational work.

On Bank of America's shipment path at the companies' own duty cycles, robot labor reaches 1% of US hours worked in 2028 and 10% in 2032. The house does not run on that path. Its volume path is built from three forecasters, Omdia's 2025 count, Goldman's 2030 base case and the median of the three 2035 forecasts, and it matches, almost exactly, what the seven announced production lines add up to. On that path, with two shifts, 85% task success, one supervisor per ten robots, and half of world shipments landing in the United States, robot labor reaches 1% of US hours around 2032 and about 5% by 2035; 10% does not arrive inside the window. On the autonomous-vehicle precedent, where reliability took a decade of public measurement before cost followed, it stays below 1% through 2035.

2. Productivity: the house base case adds about a quarter of a point a year to productivity growth between 2028 and 2035, through the cost-saving channel alone. It coincides in size with Penn Wharton's peak AI contribution of 0.2 points in 2032, which, like Goldman's 1.5 points, is a whole-economy AI estimate rather than a robot one. A path on which task success and supervision improve in a straight line from today's autonomous-vehicle values to the industry's by 2035 gives 0.3 points a year. The industry's own volume and duty cycle imply 3 points a year, twice Goldman. The autonomous-vehicle precedent implies Acemoglu's near zero. The spread is the honest state of the forecast, and most of it is volume: the same duty cycle on the industry's path and on the consensus path is 58% against 16% of US hours in 2035.

3. Why success and supervision decide it. At 99% task success with no supervisor, a \$20,000 robot delivers an hour for \$3.78 and a \$95,000 robot for \$6.22. At 70% success with one supervisor per three robots, the same machines cost \$27 and \$31 an hour, above the wage. Above 0.52 humans per robot the supervisor alone costs the median wage at any price. A robot that needs a

supervisor per three is a productivity tool for the supervisor, not a labor replacement, and the productivity effect in answer two scales with exactly these two numbers, which is why the note tracks them; neither has been published by any company in either country.

4. Confidence: five disclosures, none of which exists. Delivered hours per robot per year in a customer facility, the supervision ratio including remote operators, the task success rate achieved rather than targeted, the US share of global shipments, and service life in hours. The autonomous-vehicle industry published its version of the first three, under a state regulator, every year from 2015; the humanoid industry, in both countries, publishes none. Until at least two of the five exist from an audited source, the note's confidence in its own base case is low, and it is labeled that way.

Where the capital went, as context. Five hyperscalers spent \$181.5 billion on property and equipment in the June 2026 quarter, \$1.13 trillion since 2023. Productivity grew 2.2% over the four quarters to June and 2.5% a year over the three years since mid-2023, half a point above its 2000-to-2019 average, and the Fed's own decompositions attribute the pickup mostly to other things. Two of Nordhaus's six singularity tests print, the capital share and productivity acceleration, and one has reversed: the price of the capital good is rising relative to output. That is the base rate the four answers sit on.

Fact versus opinion. Every capex, revenue, productivity, price and shipment figure here is a filed, official or exchange-reported statistic, with two exceptions the risk factors name. Every robot cost input is a dated company claim or a named bank's estimate. The timing and productivity scenarios are the house's arithmetic on those inputs, and the base case is the house's judgment, stated with its assumptions so it can be scored.

⊕ THE OTHER SIDE FIRST

The abundance case in its strongest form

The strongest version of the case is not enthusiasm. It is five claims, each with evidence behind it.

1. **The productivity lag is the base rate for a general-purpose technology.**
Solow saw the computer age everywhere but in the productivity statistics in 1987; Oliner and Sichel found information technology explained two-

thirds of the late-1990s step-up thirteen years later. Brynjolfsson, Rock and Syverson's productivity J-curve shows measured productivity is biased down during the investment phase because complementary investments are expensed. Goldman's 1.5-point productivity boom is scheduled to begin when half of businesses have adopted; the Census Bureau's rate is 22.4%. A ledger showing capex up and productivity flat in year three is what the theory predicts.

2. **The cost of intelligence has fallen faster than any input in history.** The price of GPT-4-level performance fell 78 times in twenty-one months and the frontier still deflates four to five times a year. Nothing physical has ever done that.
3. **Volume is real and the price curve is running, in China.** Shipments rose roughly sixfold in 2025 to between 13,000 and 18,000 units depending on the counter, all six of the top vendors are Chinese, AgiBot went from its five-thousandth unit to its ten-thousandth in three months, UBTech has capacity for 6,000 a year and a 5,000-unit target for 2026, and Unitree sells a humanoid at retail for \$13,500 on a gross margin near 60% across its core businesses. The state has named embodied intelligence in the Government Work Report and the Fifteenth Five-Year Plan, backed robotics broadly with a trillion-yuan guidance fund, and some provinces and cities subsidize purchases up to 30%. Domestic reducers and screws cost half of Japanese and German equivalents. Every cost curve that mattered looked like this before volume, and Bank of America's \$13,000 to \$17,000 target requires a 21% learning rate, which is the solar-module and battery-cell rate. The bank's volume path may also be right: 2025 arrived at six times 2024, and no count exists to tell which path is running until the 2027 and 2028 shipment counts land; the choice of path is a judgment the note makes, not a test it runs.
4. **Success and supervision are software problems, and software is the input in claim two.** Figure's robot ran ten-hour shifts at BMW for ten months with a 99% target; a Chinese vendor's robot has run "24/7" on a CATL battery line since March; vision-language-action models are improving on the same curve as the token price. The reliability variables this note calls unpublished are the ones the software curve is aimed at.
5. **What this note cannot test.** Whether the current adoption rate is the start of an S-curve or its plateau; whether the intangible investment the J-curve

requires is happening; and, decisively, whether the software claim in argument four is true, because no series exists to test it. Of the five, the note tests 1 against the official decompositions, 2 directly, 3 with Wright's law and the shipment data, and scores 4 by showing what turns on it. It cannot refute 1, 2 or 4 and does not claim to.

The four answers must beat this case, not a weaker one.

⊖ ENTRY ONE · WHEN

The price is already there, the volume is in China, the hours are nowhere

The question has three parts, and it is worth saying why, because the popular argument runs them together. Mass usage needs a robot cheaper than a worker, enough robots, and robots that deliver hours. The first is a price; the second is a shipment count; the third is a reliability record. The evidence for each is different, and so is the answer.

The price is already there. Agility Robotics' June 2026 investor deck, filed with the SEC, is the only humanoid cost structure on the public record: a \$125,000 bill of materials, about 31,000 hours of service life over five years, \$15,000 a year of software and maintenance and \$15,000 to deploy. Apply that structure to every unit cost in circulation and divide by the hours the robot delivers, in dollars per delivered hour:

UNIT COST AND SOURCE	ONE SHIFT, NO SUPERVISION	TWO SHIFTS, ONE SUPERVISOR PER TEN	TWO SHIFTS, ONE PER THREE
\$20,000 Tesla's target at a million units	6.38	8.49	19.31
\$58,000 2025 average, Wood Mackenzie	7.61	9.73	20.55
\$95,000 pilot-stage bill of materials, Bank of America	8.82	10.93	21.76
\$125,000 Agility	9.80	11.91	22.74
\$200,000 Morgan Stanley, 2024	12.24	14.35	25.18

Every cell in the first two columns is below the \$22.59 production-worker wage. Energy is under five cents an hour at the industrial rate. Solving the other way, the unit cost at which a robot-hour equals the median wage is \$577,000 on one shift with no supervision and \$100,000 with a supervisor per three. Scale will lower the price further, at learning rates the banks' own targets put at the solar-and-battery slope, and that is beside the point: price has not been the gate since the first list price.

The volume is in China. This is where the robot is, and the note would be incomplete without saying so. In 2025 Chinese vendors shipped, by Omdia's count, 13,318 humanoids, up roughly sixfold on growth of 480%; IDC counts about 18,000; MERICS puts Chinese production at 12,800 units, about 90% of the world's. AgiBot shipped 5,168 and Unitree more than 5,500 by its own statement; UBTech sold 1,079 full-size industrial units for ¥821 million, with orders above ¥800 million and capacity above 6,000 a year. Unitree's average humanoid price fell from ¥593,000 in 2023 to ¥261,000 in 2024 to ¥168,000 in the first three quarters of 2025, its gross margin rose to 60%, and it priced an IPO in August 2026 to raise ¥4.2 billion, half of it for model research. Domestic harmonic reducers cost 40% to 60% of the Japanese incumbent's and roller screws half of the German one's. The state program is explicit: the 2023 MIIT guidance set batch production and a preliminary innovation system for 2025 and integration into supply chains at scale for 2027; Beijing and Shanghai each have \$1.4 billion funds; a national guidance fund and three regional funds together allocate a trillion yuan to robotics companies over twenty years.

The public spectacle is also real, and it is evidence of a specific kind. Sixteen Unitree H1s danced on the Spring Festival Gala to more than a billion viewers. Twenty-one humanoids entered the Beijing half-marathon in April 2025; four finished inside the four-hour cutoff, the winner in 2:40 with three battery swaps, and most were remote-controlled with operators running alongside. A year later more than 300 entered, the autonomous winner ran 50:26, and 40% navigated on their own. The World Humanoid Robot Games in August 2025 drew 280 teams from 16 countries. The televised robot boxing is fought by human operator teams. These events demonstrate capability, cost and national intent. They do not demonstrate delivered hours, and the marathon's own progression, from mostly remote-controlled to 40% autonomous in one year, is the reliability curve the note is looking for, published by a race organizer because no company publishes it.

What the Chinese record says about usage is more precise than the events suggest. Unitree's prospectus coverage puts 74% of its humanoid sales with research and education customers, 17% commercial and consumer, 9% industrial, and 3% to 4% in actual operational tasks. IDC's largest application category for 2025 shipments is entertainment and performance; industrial work is fifth. Among UBTech's largest orders are ¥566 million of robots to three state data-collection centers, of which about forty have been announced, where nearly a hundred humanoids at one site practice folding clothes and wiping tables hundreds of times a day to generate training data. TrendForce's caution is that reported demand mixes procurement, preorders and expressions of intent and remains "some distance from large-scale, standardized repeat purchases." China has answered the volume question and the price question. It has not answered the hours question, and an independent audit of UBTech, AgiBot and Unitree factory deployments in July 2026 found that none publishes fleet size, autonomous cycle counts, intervention rates or shift schedules.

The hours are nowhere. The entire disclosed base of delivered humanoid hours in real customer settings, anywhere, is Agility's 65,000 across nine facilities and Figure's 1,250 at BMW, about seven and a half robot-years in total. Tesla told shareholders in January 2026 that Optimus was not in use in its factories in a material way and in July that production had not started. The Chinese vendors, with a hundred times the units, disclose duty-cycle claims, a line running "24/7" at CATL that a second report pairs with an eight-hour battery, but no logged hours, fleet size or intervention rate.

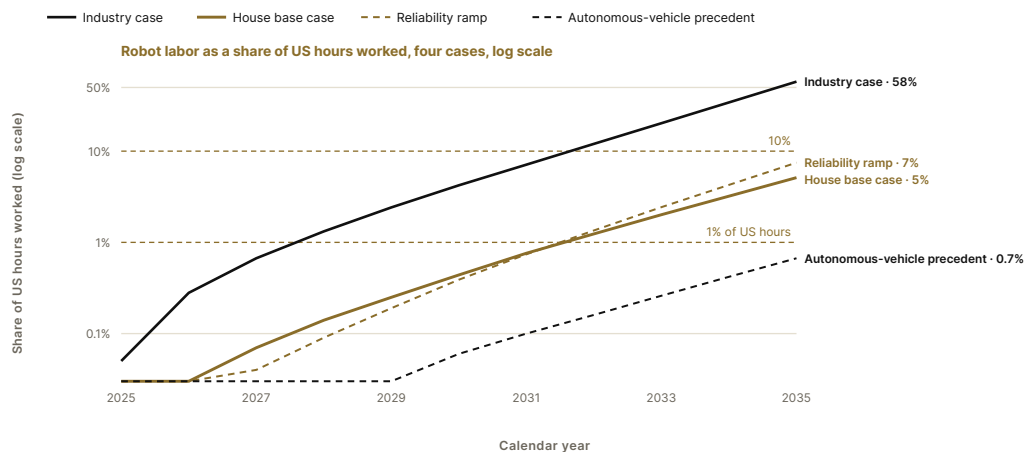
So, when. The timing question therefore reduces to three assumptions: how many robots are built, how many hours each delivers, and where they go. On the first, the note does not rely on any single forecaster. Bank of America's path, 20,000 units in 2025 to 10 million a year by 2035, is the high case.

The house's volume path. The house path runs through three anchors: Omdia's 2025 count of 13,318, Goldman's 2030 base case of 250,000, and 2.6 million in 2035, which is Omdia's forecast and the median of the three on record, Goldman at 1.4 million, Omdia, and Bank of America at 10 million. As a check, the seven production lines the vendors have announced, UBTech, AgiBot, Unitree, Figure, Agility, Boston Dynamics and a Tesla line at 5% utilization, add up to about 140,000 units a year in 2027 and, grown at 40% a year, to 6.9 million cumulative units by 2035, against 6.8 million on the house path. The two were built independently and agree on the total; year by year the capacity check runs ahead of the consensus path, at roughly 190,000 units in 2028 and 380,000 in

2030 against 77,000 and 250,000, closer to the bank's path in the near years, which is where the first-percent date is set. The 40% ramp after 2027 is the house's assumption. The note runs four cases and labels the second as the house's.

CASE AND ASSUMPTIONS	VOLUME PATH	1% OF US		SHARE, 2035	\$ PER DELIVERED HOUR, 2030
		HOURS	10%		
Industry · companies' duty cycle, 20 h/day, 99% success, no supervision, every shipment US	Bank of America, 10 million a year by 2035	2028	2032	58%	\$3
House base · two shifts, 85% success, one supervisor per ten, half of shipments US, four-year life, all held constant	Consensus, 2.6 million a year by 2035	2032	after 2035	5%	\$10
Reliability ramp · autonomous-vehicle values in 2025 improving in a straight line to the industry's by 2035; half of shipments US, four-year life	Consensus	2032	after 2035	7%	\$15
Autonomous-vehicle precedent · one shift, 70% success, one per three, a third of shipments US, held constant	Goldman low, 1.4 million a year by 2035	after 2035	after 2035	0.7%	\$36

EXHIBIT 1 Four timing cases on one axis



How to read: the horizontal axis is the calendar year; the vertical axis is the share of all hours Americans work that humanoid robots deliver, on a log scale so that 0.1%, 1% and 10% sit equal distances apart. Each line is one of the four cases in the table above. They differ in two things: the volume path, Bank of America's shipments for the industry case and the consensus path for the other three, and the reliability assumptions, the duty cycle, task success and supervision each case holds or ramps. Gold hairlines mark 1% and 10%.

Only the last row's cost, \$36 an hour, is above the wage. The house base case rests on four judgments, stated so they can be scored. The consensus volume path is the first, and it is the one that matters most: on the same duty cycle, Bank of America's path gives 58% of US hours in 2035 and the consensus path 16%. Half of world shipments reaching the United States assumes the announced American lines, Tesla's million-unit Fremont line, Figure's 12,000-a-year plant and Agility's 10,000, reach volume by the early 2030s; today the US share is under a tenth. Eighty-five percent success and one supervisor per ten is the house's guess at where the reliability curve sits, on average, across the ramp: the Beijing marathon went from mostly remote-controlled to 40% autonomous in a year, and the autonomous-vehicle record improved fivefold in three years once measurement began. Two shifts rather than the companies' three assumes charging, maintenance and the customer's own schedule take a third of the day, as Agility's own deck assumes for its current model. On those judgments robot labor is about 1% of US hours in 2032 and 5% in 2035, four years behind the industry case at the first percent; if the autonomous-vehicle record is the guide it stays below 1% through 2035.

The first, second and fourth rows hold task success, supervision and duty cycle constant through 2035; none of them models the improvement the proponents'

fourth claim predicts, and that is a limit of the frame, not a finding, because the house has no series on which to fit a curve. The third row is the proponents' claim expressed inside the frame: success, supervision and duty cycle start at the autonomous-vehicle values in 2025 and improve in a straight line to the industry's by 2035. It reaches 1% of US hours in the same year as the house base case, 2032, and ends 2035 higher, at 7%, because the improvement arrives late in the decade when the installed base is largest. The house base case is the house's guess at the midpoint of that ramp, and the ramp row is what a proponent would say the true path looks like.

66,000

Every delivered humanoid labor-hour any company has disclosed from a real customer facility, in either country, Agility's 65,000 and Figure's 1,250: about seven and a half robot-years, against a service-life assumption of 31,000 hours per robot and a Chinese installed base a hundred times larger that discloses none.

Σ ENTRY TWO · PRODUCTIVITY

A quarter of a point a year on the house path, three points on the industry's

The method is stated in one sentence so it can be argued with. The saving from replacing a worker-hour with a robot-hour is the labor share of income, half of gross domestic income today, times the share of hours replaced, times one minus the robot's cost per delivered hour as a fraction of the \$46.60 fully loaded cost of a human hour. Spread over the 2028-to-2035 ramp, that is an addition to productivity growth. It counts only the cost-saving channel; it does not count the output that cheaper labor would call into existence, which is the channel the proponents would add and the one the note has no basis to size.

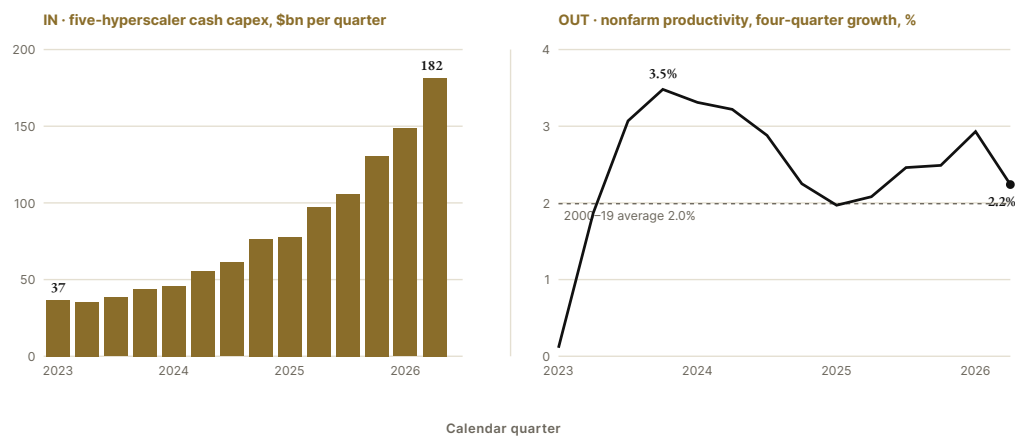
CASE	HOURS REPLACED, 2035	SAVING ON A LOADED HUMAN HOUR	OUTPUT LEVEL EFFECT, 2035	ADDED PRODUCTIVITY GROWTH A YEAR, 2028-35
Industry (Bank of America volume)	58%	93%	27%	3.0 points
House base (consensus volume)	5%	78%	2%	0.25 points
Reliability ramp (consensus volume)	8%	68%	2.6%	0.32 points
Autonomous-vehicle precedent (Goldman-low volume)	0.7%	24%	0.1%	0.01 points

The spread is the point, and most of it is volume. The industry's own inputs imply a productivity boom twice the size of Goldman's, which would be visible in the national accounts within two years of starting and which Goldman itself does not forecast. The autonomous-vehicle precedent implies Acemoglu's ceiling of 0.66% of total factor productivity over ten years, an effect too small to detect. The house base case, a quarter of a point a year for eight years, coincides in size with Penn Wharton's peak AI contribution, a whole-economy estimate rather than a robot one, and would be at the edge of detection in a series that is revised by more than that. The house's productivity answer is therefore modest for this decade, and it is modest because the consensus volume path is about a quarter of the industry's, not because the robot is expensive or the hours are few once it is built. Every part of it depends on the two numbers in question three.

The base rate it sits on. The capital is already in. Five hyperscalers spent \$36.6 billion on property and equipment in the March 2023 quarter and \$181.5 billion in the June 2026 quarter, 2.2% of GDP at an annual rate; NVIDIA's data-center revenue went from \$4.3 billion to \$89.0 billion a quarter. Against that, nonfarm business productivity grew 2.2% over the four quarters to June 2026 and 2.5% a year over the three years since mid-2023, half a point above its 2000-to-2019 average of 2.0% and inside the range it printed in 2023 before the buildout was large. The Kansas City Fed's industry decomposition finds the pickup concentrated in four industries and AI explaining little of it; the San Francisco Fed's regime model puts the probability of a high-productivity regime at 57% on labor productivity and 21% on total factor productivity, and reads the gap as

better tools rather than more efficiency. Nordhaus's six tests for an approaching acceleration, re-run on 2023-to-2026 data, print two: the capital share of income is rising at three times its old trend, and productivity growth clears his bar by exactly half a point. One test has reversed. The BEA's price index for information-processing equipment and software, which fell 7.8% a year relative to output for two decades, rose 1% a year in the AI window: the core capital good of this buildout is getting dearer, which is what a supply-constrained input looks like. None of this contradicts the base case; it is what year three of the J-curve looks like. It does mean the productivity effect in the table has not begun, and the note's forecast is that it begins around 2028, not that it is under way.

EXHIBIT 2 The base rate: capital in, output out



How to read: both panels run over the same quarters, 2023 to mid-2026, so the input can be read against the output. Left, each bar is the cash Microsoft, Alphabet, Amazon, Meta and Oracle together spent on property and equipment in one quarter, in billions of dollars. Right, the line is how much output per hour in the nonfarm business sector grew over the prior four quarters, in percent, and the dashed line is its 2000 to 2019 average. The right-hand series is the one the house's July note on this capex, *The AI Capex Payback Clock*, used as its output test.

ENTRY THREE · THE TWO NUMBERS

Why task success and supervision decide everything

The two variables that set the timing in question one and the productivity effect in question two are the same two, and they are the two the industry does not report. Before asking what disclosure would settle the forecast, the note shows how much turns on them.

UNIT COST	TASK SUCCESS	NO SUPERVISOR	ONE PER TEN	ONE PER THREE	0.52 PER ROBOT
\$20,000	99%	3.78	8.49	19.31	28.26
\$20,000	85%	4.40	9.89	22.50	32.91
\$20,000	70%	5.35	12.00	27.32	39.96
\$95,000	99%	6.22	10.93	21.76	30.70
\$95,000	85%	7.25	12.73	25.34	35.76
\$95,000	70%	8.80	15.46	30.77	43.42

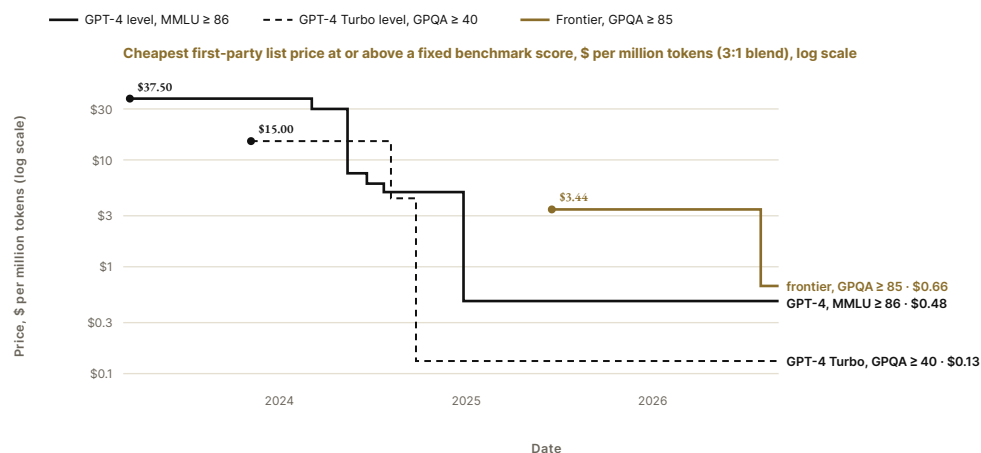
The grid is in dollars per delivered hour, two shifts, 31,000-hour life, against a median wage of \$24.51. For comparison, the one input that has followed a software curve, the price of a fixed unit of intelligence, fell 78 times in twenty-one months before finding a floor. Read across a row and the price of the robot barely matters: a \$20,000 robot and a \$95,000 robot are \$2.44 to \$3.46 apart at any given success and supervision, the gap widening as success falls. Read down a column and supervision matters more than anything: a supervisor per ten robots adds \$4.71 to every delivered hour at the loaded human rate and 99% success, a supervisor per three adds \$15.53, and at 0.52 humans per robot the supervisor alone costs the median wage. Task success multiplies both, because the supervisor is paid for the failed hours too. A robot at 70% success with one supervisor per three costs more than the worker it replaces at every price on the ladder. That is the whole economic case in one table, and it is also the productivity case: a machine that needs a human for every three of it is a tool that makes the human more productive, which is Acemoglu's world, not a labor supply, which is the industry's.

What is known about the two numbers is thin and the note lists it. Figure's target at BMW was 99% per shift and it reports a sevenfold improvement over an undisclosed base. Agility and Figure claim no teleoperation; 1X's chief executive says much of the early work will be done by teleoperators; Tesla's 2024 demonstration used remote operators, per Bloomberg; the Chinese marathon went from mostly remote-controlled to 40% autonomous in a year, and the televised fights are operator-controlled. No company on either side of the Pacific has published an achieved success rate or a supervision ratio.

The proponents' answer, and the one precedent. The proponents' reply is that success and supervision are software, and software is the input whose price this

note finds falling four to five times a year at the frontier and eighty-fold at fixed capability before hitting a floor. That may be right, and the note has no series to test it. The one machine that published its reliability year by year while its cost curve ran is the autonomous vehicle, because California's regulator required it. Waymo reported about 5,600 miles between safety-driver interventions in 2017, about 30,000 in 2020 and 17,300 in 2023 as its test mix shifted, with driverless miles rising from 52,000 in 2022 to 3.9 million in 2025. It began driverless public service in 2020, eleven years after the project started, reports 220 million rider-only miles through March 2026, and describes its sixth-generation hardware as cutting cost with fewer than half the cameras. Reliability was measured publicly for a decade, the delivered-miles base was built, and the cost cut came after. That sequence is the autonomous-vehicle case in the timing table, and the reason the house base case sits closer to it than to the industry's.

EXHIBIT 3 The cost of a fixed unit of intelligence



How to read: the horizontal axis is the date; the vertical axis is the price of a million tokens from the cheapest model that clears a fixed benchmark bar, on a log scale. Each step down is a cheaper model clearing the bar, and a flat stretch is a floor no vendor has undercut. The lines differ only in the bar: GPT-4's level on one benchmark, GPT-4 Turbo's level on another, and the frontier. This is the software curve the proponents say task success and supervision will ride.

ENTRY FOUR · CONFIDENCE

Five disclosures, none of which exists

The note's confidence in its base case is low, and the reason is specific. Five disclosures would make the timing and productivity forecasts in questions one

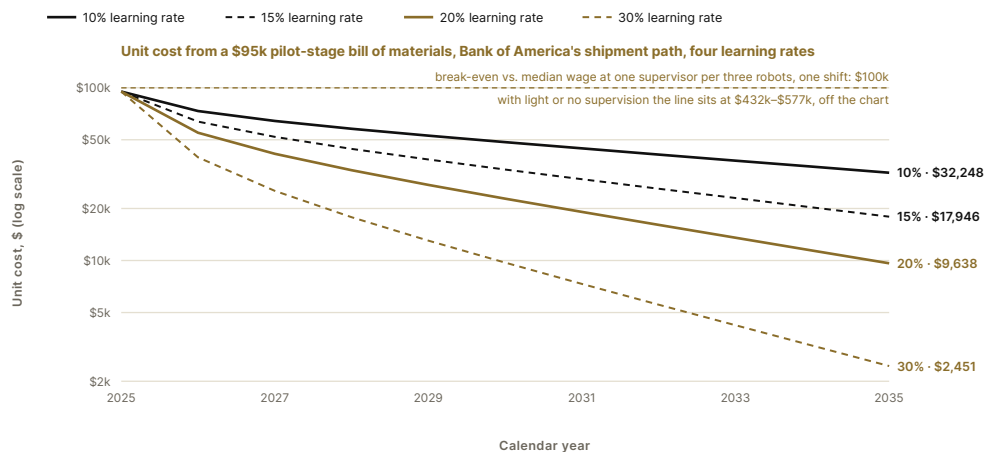
and two trustworthy, and none exists from any company in any country:

1. Delivered hours per robot per year in a customer facility, from an auditor or a customer rather than the vendor.
2. The supervision ratio, humans per robot, including remote operators.
3. The task success rate achieved per shift, not the target.
4. The US share of global humanoid shipments.
5. Service life in hours, at retirement rather than per battery.

The first document likely to carry the first three is Agility's registration statement for its SPAC merger, which will have to describe operating data under securities law. Unitree's exchange filings will carry units and prices but, on the prospectus coverage so far, not hours. The precedent for what a disclosure regime looks like is the California DMV's annual disengagement report, which the autonomous-vehicle industry filed from 2015 and which made the reliability curve public a decade before the cost cut. Until two of the five exist from an audited source, the house base case is a judgment about the shape of a curve nobody has published, and the note re-scores it as each disclosure lands.

What would move the base case. The scale objection is that the robot is pre-volume and the cost curve has not run. Bank of America's path is ten cumulative doublings by 2035 and the consensus path eight; at the solar rate of 20% per doubling a \$95,000 unit is \$9,600 on the first and \$14,500 on the second, and the \$13,000 to \$17,000 target requires 20.7% on the bank's own volume. But every learning-rate path starts below break-even, so scale moves the price, not the date. What moves the date is the reliability curve, and the only precedent for it is the decade the autonomous vehicle needed.

EXHIBIT 4 Wright's law on the industry's volume path



How to read: the horizontal axis is the year; the vertical axis is the cost of one humanoid in dollars, on a log scale, starting from a \$95,000 pilot-stage bill of materials. Each line lets that cost fall as cumulative production doubles along Bank of America's shipment path, and the lines differ only in the learning rate, the percentage the cost falls with each doubling. The gold dashed line is the unit cost at which a robot-hour equals the median wage with one supervisor per three robots; every line starts below it. On the consensus path, with two fewer doublings, each line lands about half again higher by 2035.

THE BALANCE

What the four answers sum to

The robot exists as a product and a price, in China first, at a cost per hour already below the wage. It does not yet exist as a labor supply anywhere, because the two numbers that turn a product into a labor supply, task success and supervision, are unpublished, and the one precedent for publishing them says the curve takes a decade to run. The house's best guess is that robot labor reaches 1% of US hours around 2032 and about 5% by 2035, adding a quarter of a point a year to productivity growth from 2028; the industry's case, on Bank of America's volume, is four years earlier at the first percent and reaches 10% in 2032, and the autonomous-vehicle case stays below 1% through 2035. The capital for the cognitive half of that forecast is already in the ground and has not yet shown up in output, which is the base rate and not a verdict.

Next: Robots, AI, Abundance, and the Debt

Part 2 takes this note's timing cases as given, the industry's and the house's, and follows the robot-hour into the national accounts. If everything a factory makes gets cheaper, does the national debt get paid down, does a basic income get funded, and who collects the rent? The debt question turns on one variable that has nothing to do with robots, the other two on who owns what, and Part 2 tests all three against the one franchise that tried to write money out of its economy on screen.

≡ WHAT THIS LEDGER CANNOT SEE

Risk factors

- **Volume is the largest single assumption and no forecaster carries it alone.** The industry case uses Bank of America's 10 million units a year by 2035; the house path runs through Omdia, Goldman and the median 2035 forecast and is cross-checked against announced capacity, but the 2025 counts it starts from already differ by a third (Omdia 13,318, IDC 18,000, MERICS 12,800), TrendForce warns that reported demand mixes orders, preorders and intent, and the capacity check assumes lines that are announced, not built. A slower path moves every date later; the bank's path moves them four years earlier.
- **The house base case is a judgment,** and its four assumptions are stated so they can be wrong in public: the volume path, US share of shipments, the reliability level, and the duty cycle. The autonomous-vehicle precedent is one precedent.
- **The cost model applies Agility's structure to every robot.** The 31,000-hour life is the company's illustrative figure; Tesla's 500-watt draw is used for all; maintenance is Agility's. It is a stress on the inputs, not anyone's cost estimate.
- **The productivity method counts one channel.** Output expansion from cheaper labor is not modeled and would raise every row; adjustment costs and displacement are not modeled and would lower them.
- **Chinese figures come through exchange-filing coverage and state media,** not the filings themselves. Unitree's prospectus and UBTech's results

announcement were read through Caixin, 36Kr, Gasgoo and SCMP; the unit split behind "more than 5,500" is unverified; net profit for 2025 is reported at two different figures by two outlets. The customer-mix percentages are from prospectus coverage.

- **The capex series undercounts**, excluding finance-lease principal and the private neoclouds; the productivity series is revised, and the half-point acceleration is inside the revision range. Nordhaus tests four and five run on data ending in 2024.
- **Waymo's miles between interventions are compiled from press summaries of the California DMV releases**, and the 2020 figure appears in two slightly different secondary values; the note cites the direction and the order of magnitude.

↪ REVERSAL

What would change our read

1. **An audited disclosure of delivered hours, supervision ratio and success rate from any deployment, in either country, showing 5,000 or more hours per robot per year, 20,000 over the four-year life the base case assumes, at one supervisor per ten or better.** That is the base case's own assumption printed by a third party, and it would move the house toward the industry case by about two years. The mirror holds at the same distance: a filing showing under 1,500 hours per robot-year, 6,000 over the same life and clearly below the 8,000 to 10,000 a \$95,000 unit needs to break even, or supervision worse than one per three, moves the house toward the autonomous-vehicle case by about two years.
2. **The Beijing marathon or the Humanoid Games reaching 90% autonomous entrants**, on the organizers' own count. The reliability curve the note infers from 40% in 2026 would then be running faster than the autonomous-vehicle precedent.
3. **Global shipments exceeding 250,000 units before 2030, or US shipments exceeding 100,000 in a calendar year before 2030**, on filed or counted data. The first is the consensus path running at the bank's pace, the second is the base case's US-share assumption arriving early; either moves the house toward the industry case by two years. The mirror holds at the same distance: global shipments below 100,000 in 2030, or a US

share still under a fifth of the world's on counted data in 2030, moves the house toward the autonomous-vehicle case by two years.

4. Productivity growth above 3% for four consecutive quarters with the San Francisco Fed's total-factor-productivity regime probability above 50%.

That would mean the cognitive half of the buildout was already producing output, and the productivity forecast in question two would need to start earlier.

5. The Chinese counters converging below 10,000 units for 2025 or 2026 shipments falling short of the vendors' targets by half, which would put

the volume path, and every date in the timing table, in doubt in the industry's disfavor.

6. The token price floor at fixed 2024 capability falling below \$0.10 per million, meaning the software curve the proponents rely on for success and supervision had not stopped, which would move the house toward the industry case on question three.

Tripwires, re-scored on release: Agility's S-4; Unitree and UBTech exchange filings (semi-annual); Omdia, IDC and TrendForce shipment counts (annual); the Beijing half-marathon and Humanoid Games autonomy counts (annual); BLS Productivity and Costs (quarterly); hyperscaler 10-Qs and NVIDIA 8-Ks (each cycle); Census BTOS (biweekly); first-party token price lists (monthly).

APPENDIX

Sources & method

Robot-hour cost and timing: Agility Robotics investor presentation and press release, Churchill Capital Corp XI 8-K exhibits 99.1 and 99.2 (June 2026); Tesla earnings-call transcripts Q2 2025 to Q2 2026 and the 2025 shareholder meeting; Figure AI blog posts (Nov 2024 to Nov 2025) and press on the May 2026 continuous-operation test; BMW Group press release 2026-06-25; 1X, Boston Dynamics and Apptronik disclosures; Goldman Sachs (Feb 2024), Morgan Stanley (May 2025), Bank of America Institute (Apr 2025, Mar 2026), Citi GPS (Dec 2024), Wood Mackenzie via CNBC (Aug 2026); IFR World Robotics 2025. All inputs archived with verbatim quotes and claim-owner tags in `data/sources/robot-hour/`; model in `data/robot_hour_model.py`; shipment paths in `data/volume_paths.py` (Omdia via SCMP, IDC via CGTN, MERICS, Goldman Sachs Feb 2024, Bank of America Mar 2026, and the announced lines of UBTech, AgiBot, Unitree, Figure, Agility, Boston Dynamics and Tesla); timing and productivity scenarios in `data/mass_usage.py`; Wright's-law paths in `data/wrights_law.py`; US hours from BLS payrolls and average weekly hours. Wage comparators: BLS OEWS May 2025 and ECEC 2026 Q1 via the BLS API; electricity: EIA Electric Power Monthly, June 2026.

China: Unitree IPO prospectus coverage (Gasgoo, 163.com, 36Kr, PANews, Sina; company statement of 2026-01-22); UBTech 2025 results coverage (Gasgoo, Humanoids Daily) and Walker

S2 press release (Nov 2025); AgiBot company releases (Jan and Mar 2026); Leju prospectus coverage; Omdia via SCMP (Jan 2026); IDC via CGTN (Jan 2026); MERICS embodied-AI report (Apr 2026); TrendForce (Aug 2026); USCC translation of the MIIT 2023 Guiding Opinions (Oct 2024); gov.cn and Yicai on the 2025 programs; TechCrunch and PBS on the 2025 and 2026 Beijing half-marathons; Global Times on the World Humanoid Robot Games and the CMG Mecha Fighting Series; SCMP on the Spring Festival Gala; Gasgoo on Galbot at CATL; Rest of World on the data-collection centers; TechniaHQ deployment audit (Jul 2026); 36Kr on component costs. 29 files in [data/sources/china/](#).

Capex, productivity and the Nordhaus tests: SEC XBRL company facts for Microsoft, Alphabet, Amazon, Meta, Oracle and NVIDIA; NVIDIA 8-K earnings exhibits; cross-checked against Federal Reserve FEDS Note 2026-07-17 (matches to rounding for the four large names); FRED/BEA/BLS series listed in [data/fred/_manifest.json](#); Nordhaus, NBER w21547, re-run per [data/nordhaus_tests.py](#); FRBSF Economic Letters (Feb and May 2026); Kansas City Fed (Dec 2024, Feb 2026); BLS Productivity and Costs Q2 2026 revised; Census BTOS through 2026-08-09; Solow, Oliner and Sichel, Brynjolfsson-Rock-Syverson, Acemoglu (NBER 32487), Goldman Sachs (Briggs and Kodnani, 2023).

Cost of intelligence and the reliability precedent: first-party vendor price lists 2023 to 2026 with Wayback snapshots; Epoch AI Benchmarking Hub and inference-price analysis; a16z; 63 dated rows in [data/intelligence_cost.csv](#), curve in [data/intelligence_cost.py](#). Learning rates: Lafond et al. 2018, Our World in Data, BloombergNEF, Ziegler and Trancik, Nykvist and Nilsson, Barwick et al., BCG 2015, Nordhaus 2007; adoption: Pew, Census P23-208, McGrath 2013; reliability: California DMV disengagement reports 2018 to 2026 with EE Times, The Robot Report and The Last Driver License Holder compilations; Waymo Safety Impact hub and blog (Feb 2026); The Robot Report on robot-cell reliability. [data/sources/cost-curves/](#).

Limitations: the timing and productivity cases are arithmetic on the industry's volume forecast and the house's stated assumptions, not observed adoption; the cost model is a stress test on one company's structure; Chinese figures are exchange-filing coverage rather than filings; every conclusion is dated to the vintages above and re-scores on the tripwires listed.

DISCLOSURES

Disclosures

Information only. TON618 Capital. This note is for information purposes only. Nothing here is an offer to sell or a solicitation of an offer to buy any security, fund interest, or digital asset, and nothing here is personalized investment advice or a recommendation regarding any instrument.

Publisher's exclusion. All research is published solely as general, impersonal information of regular circulation. It is not tailored to the objectives or circumstances of any individual and is not issued in connection with compensation from any client. The Fund has no clients and distributes all research free of charge. On that basis it publishes in reliance on the publisher's exclusion from the definition of "investment adviser" under the Investment Advisers Act of 1940 (§202(a)(11)(D); cf. *Lowe v. SEC*, 472 U.S. 181 (1985)).

Registration & conflicts. TON618 Capital is not registered as an investment adviser or broker-dealer in any capacity. The Fund is a Bitcoin fund and may hold or transact in securities or digital assets discussed, including the technology, semiconductor and robotics names referenced in passing. Material conflicts are disclosed where they exist. The Fund receives no compensation from any party in connection with its research.

Use of AI. Artificial intelligence is used in the creation of this research. All methodology and data integrity are reviewed and approved by TON618 Capital's Chief Investment Officer, Keyth Beck, before publication; errors may nonetheless occur, and readers should verify independently.

CFA. CFA® and Chartered Financial Analyst® are registered trademarks owned by CFA Institute. References to CFA Institute standards describe the analytical framework applied and do not imply certification by, membership in, endorsement by, or affiliation with CFA Institute.

Risk & feedback. Past performance is not indicative of future results. Digital assets and equities are volatile and may result in total loss of capital. Corrections and feedback are welcome at keyth@ton618capital.com.